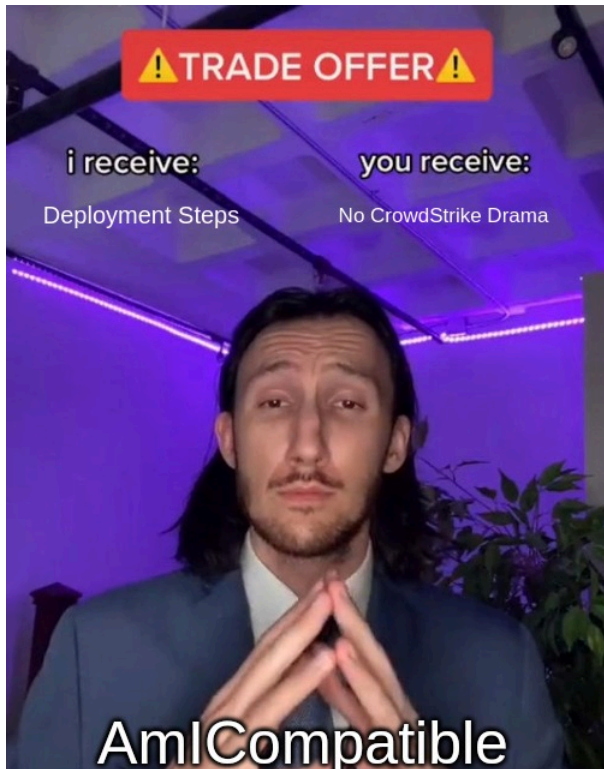


AmlCompatible

For automated cross-platform testing.

The world has lots of different Operating Systems, CPUs have different architectures. You want your app to work for everyone, but buying every single platform to test your computer on?

We don't do that here! AmlCompatible, or aic for short is capable of starting multiple virtual machines to execute your CI/CD pipeline on all those different architectures. The tool is simple and builds on other existing industry standards.



We plan to make this a single executable to which you provide the necessary flags as parameters, it might look like this:

```
aic config.aic Jenkinsfile
```

You provide a configuration file which dictates what platforms you want to test your app on, you also provide information of server addresses where the servers can be reached. The second option is your [Jenkinsfile](#) used to run your pipeline. Now, `aic` can launch the required VMs and provide you with information on how well your program did.

This app will involve a lot of scripting and know-how of DevOps tools and Infrastructure. If you have interest in those you surely have an interest in this project!

MVP

For **aic** to be working we need this:

- Work with static binaries or scripts without dependencies, it must be a single entrypoint
- Supporting Ubuntu as host OS
- Able to run multiple vms as instructed from a configuration file
- Statistics: test results and/or if deployment succeeded
 - Done by sending the logs of the tests
- *Your help!*

Future

- **Support for advanced deployments (parsing CI/CD to chef per vm)**
- Support W11, Windows Server 2022, MacOS, Debian, RHEL
- Support for multiple runners at once that sync and distribute VMS
- Support for different CPU architectures like ARM, x86 (select what servers are arm and what are X86 and we will test all vms on both architectures)
- Support for cloud providers
- More statistics (cpu usage, ram usage, test time,...)
- Support for virtualization apps (test my .exe app in bottles)

Tech Stack:

- CI/CD (Jenkins)
- Virtualization (VMware/VirtualBox/...)
- Linux & Windows
- Scripting (Powershell/GO/Python)
- Monitoring (grafana/prometheus/ELK)
- Cloud (Azure/AWS) + Terraform ? (need user api key)

-Jarno & Luca